Wavelet Analysis

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Introduction

Wavelet analysis decomposes a time series into components localised in both time and frequency simultaneously. Unlike the Fourier transform, which assumes stationarity and produces a single spectrum for the entire series, wavelets capture cyclic components that appear, shift, or disappear over time — essential for non-stationary signals such as EEG recordings, climate proxies, or any series whose periodic content evolves. The output is a scalogram, a 2D map of power across time and frequency that reveals the time-varying structure that Fourier methods miss.

Prerequisites

A working understanding of spectral analysis and Fourier transforms, the concept of time-frequency localisation, and basic familiarity with the trade-off between time and frequency resolution.

Theory

The continuous wavelet transform (CWT) convolves the signal with scaled and shifted copies of a mother wavelet ψ:

$$W(a, b) = \int y(t)\, \bar{\psi}\left(\frac{t-b}{a}\right) \frac{\mathrm{d}t}{\sqrt{|a|}}.$$

The scale a controls the wavelet's "width" and corresponds inversely to frequency; b is the time shift. The squared magnitude $|W(a,b)|^2$ is the wavelet power at time b and scale a. Common mother wavelets include Morlet (oscillatory, good time-frequency localisation), Haar (step-function), Daubechies (orthogonal, used in compression). The scalogram visualises $|W|^2$ as a heatmap over the time-frequency plane.

The cone of influence delimits regions of the scalogram free of edge artefacts; power within the cone is reliable, power outside should be discounted.

Assumptions

Depends on the wavelet choice; minimal beyond a sufficiently long series. The discrete wavelet transform (DWT) requires series length to be a power of 2; the CWT does not.

R Implementation

```r
library(WaveletComp)

set.seed(2026)
t <- 1:500
y <- sin(2 * pi * t / 50) * (t < 250) + sin(2 * pi * t / 20) * (t >= 250) +
     rnorm(500, 0, 0.3)

df <- data.frame(date = t, x = y)
```

```r
wt <- analyze.wavelet(df, "x", loess.span = 0, dt = 1,
                      dj = 1/50, lowerPeriod = 8,
                      upperPeriod = 128, make.pval = FALSE, verbose = FALSE)

wt.image(wt, main = "Wavelet power spectrum")
```

Output & Results

`analyze.wavelet()` returns a list containing the wavelet power, phase, and statistical-significance information. `wt.image()` plots the scalogram as a 2D heatmap with the cone of influence overlaid.

Interpretation

A reporting sentence: "The wavelet power spectrum revealed a 50-time-step cycle in the first half of the series and a 20-time-step cycle in the second half — a non-stationary feature invisible to a global Fourier analysis. The cone of influence excluded the first and last 30 time points from interpretation due to edge effects." Always describe the time-frequency structure visible and acknowledge the cone of influence.

Practical Tips

- Choose the wavelet to suit the signal: Morlet for oscillatory components, Mexican-hat for localised peaks, Daubechies for compression and denoising.
- Power outside the cone of influence is unreliable; mask it or interpret with caution.
- Statistical significance via red-noise surrogates (`make.pval = TRUE` in `WaveletComp`) tests power against an AR(1) null at each scale and time.
- The discrete wavelet transform (DWT, in `wavethresh`) supports compression and denoising; CWT is for visualisation and analysis.
- For EEG, fMRI, and other neurophysiological data, wavelets are the standard time-frequency tool; specialised packages (`eegkit`, `fmri.wavelets`) extend the framework.
- For multi-channel time-frequency analysis, cross-wavelet coherence and wavelet phase quantify time-varying co-movements between series.

R Packages Used

`WaveletComp` for `analyze.wavelet()`, `wt.image()`, and biwavelet coherence; `wavelets` and `wavethresh` for discrete wavelet transforms and threshold-based denoising; `dplR` for tree-ring-style wavelet analysis; `Rwave` for additional wavelet families and inversion.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics